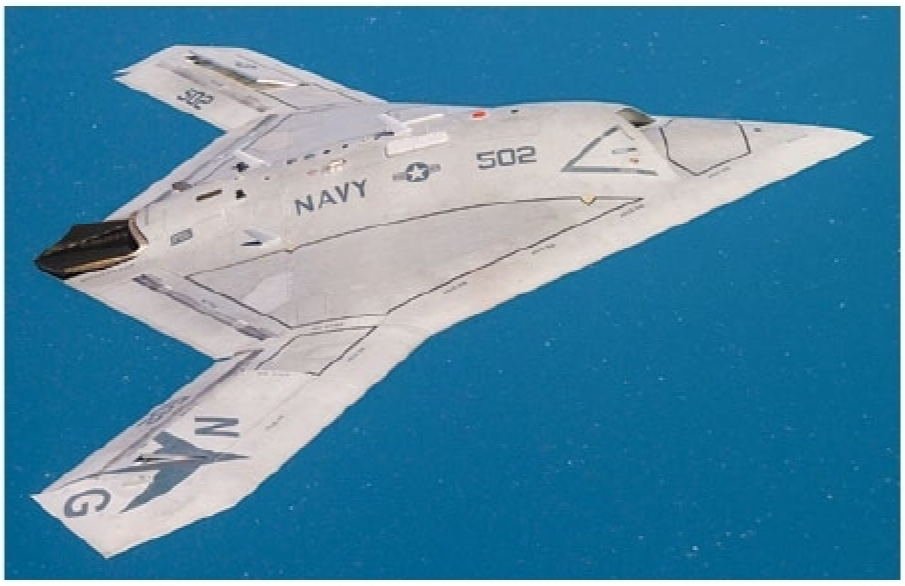
Rise of the real killer robot in the air

So far, we have seen the capabilities and applications of AI on Earth and in the oceans. What if AI is deployed in the air? AI in air space is more profound, devastating and can cause colossal damage to people. You may be hiding underground, underwater or anywhere on Earth, and AI-assisted devices can easily find you. Many countries are developing AI-assisted weapons such as aircraft, hypersonic missiles, submarines and drones. China, Israel and the US are widely recognised as the industry leaders in unmanned combat aerial vehicle (UCAV) technology.



X-47B Pegasus UCAV (Credit: Wikipedia)

On offensive systems, AI is employed to achieve a higher degree of autonomy. LAWs would include drones or UCAVs. For example, the unarmed BAE Systems Taranis drone prototype can autonomously search, identify and locate enemies. It can also defend itself against enemy aircraft.

Another example is Northrop Grumman X-47B Pegasus UCAV, which is a tailless jet-powered blended-wing-body aircraft capable of semi-autonomous operation and aerial refuelling. It is known as the best military UCAV. It can take off and land on aircraft carriers, and is set to be developed into an unmanned carrier-launched airborne surveillance and strike (UCLASS) system.

Boeing has unveiled an autonomous fighter jet that is expected to launch in 2020. The aircraft is designed to fly with crew during combat, gather intelligence, and perform early warning, surveillance and reconnaissance.

Global giants in AI arms race

While Google has refused a contract deal with US Department of Defence citing that the technology should not be used for destruction, Microsoft and Amazon have been selected by US Department of Defence as the two finalists for its US$ 10 billion Joint Enterprise Defence Infrastructure (JEDI) cloud contract.

On the other hand, China's AI industry aims to take the lead in the AI arms race. Over the forecast period of 2019 to 2025, China's defence expenditure on AI is expected to record a CAGR of 21.6 per cen, increasing from US$ 265 to reach US$ 1041.8 million by 2025. China is also developing large autonomous submarines suitable for reconnaissance, mine placement and suicide attacks against enemy vessels, slated for deployment in the early 2020s.

Many other countries like the UK and Russia are also developing autonomous weapons and drones. Experts think that the arms race between these countries will someday inevitably lead to a system of artificial general intelligence (AGI). Development of AGI could be the final arms race, after which there would be a new chapter in human history.

With the emergence of AI technology, the world is in a great dilemma. While AI brings many benefits to the society, there are also great risks involved, including ethical issues and destruction. Hardware is easily available, and software has become open source. Bad guys could use the technology for mass destruction.

Indian defence in AI space

The government of India has initiated the process of preparing Indian defence forces to use and develop AI technology within the country. Ministry of Defence has set up Defence AI Project Agency (DAIPA), an agency that lays down standards for technology development and delivery process for AI projects, establishes the standard operating procedure for these projects, formulates policy for IPR and provides for selection of strategic partners.

Indian defence forces are planning to use AI to identify and index military objects in aerial photographs or in videos. Report says that the armed forces are launching an AI engine that can further analyse satellite photos and video feeds from drones to identify images and patterns in real time.

According to reports, Army Design Bureau, a part of Make in India programme, has built a device that will warn soldiers of unusual activity across different terrains, including in high altitude areas.

Another project has been sanctioned to a DRDO-based laboratory to develop AI-based solutions to signal intelligence to enhance intelligence collation and analysis capabilities for the armed forces.

Conclusion

AI has become an essential element of defence programmes across the world. Many countries are taking part in the AI arms race, and autonomous weapons are being developed. LAWs could be the weapons of terror and destruction in future. Many countries are focusing on developing cyber warfare capabilities to defend their national infrastructure and attack against foreign adversaries in cyberspace. There are high chances of rogue states and terrorist groups breaching the cybersecurity of one or more nuclear-armed countries, and setting in motion a nuclear attack in future.

The Indian government has also initiated the process of preparing defence forces for the use of AI. Experts think that in future, wars will be fought by AI-based autonomous machines, but whether humans will still be in charge is difficult to predict right now. Hence, do not underestimate the power of AI! And let me ask you the question again: Are you afraid of AI? EFY